FILTER DESIGN TEST CASE

Multiplexer Module - n78 (3.3-3.8 GHz 5G NR) for Satellite L-Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0076
Revision:	Rev D.8
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Created Date:	2024-07-15
Last Modified:	2024-08-01
Status:	NEEDS REVIEW
Priority:	P3 - Medium

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Multiplexer Module filter targeting n78 (3.3-3.8 GHz 5G NR) for Satellite L-Band Filter. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 746.9 MHz and bandwidth of 328.3 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Satellite L-Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Multiplexer Module
- Frequency Band: n78 (3.3-3.8 GHz 5G NR)
- Application: Satellite L-Band Filter
- Substrate: 36° YX LiTaO3
- Package: CSP (Chip Scale Package)
- Design Tool: CST Studio Suite 2024
- Design Center: Osaka Design Center

This specification applies to all variants of the Multiplexer Module design targeting n78 (3.3-3.8 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 746.9 MHz
Bandwidth (3 dB): 328.3 MHz
Insertion Loss Target: <= 1.5 dB
Return Loss Target: >= 22.0 dB
Out-of-Band Rejection: >= 32.0 dB
Isolation (Duplexer): >= 39.0 dB
Q Factor: 5664
Temperature Coefficient:-6.1 ppm/°C
Die Size: 0.7 x 2.1 mm
Wafer Size: 6-inch
Number of Resonators: 11
Electrode Material: Mo

Passivation: Si3N4

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the Multiplexer Module template project.
2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 746.9 MHz, BW 328.3 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.5 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 36° YX LiTaO3).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 2241 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.5 dB
- Return loss in passband shall be >= 22.0 dB
- Out-of-band rejection at +/- 492 MHz offset >= 32.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 0.6 MHz
- Power handling shall be >= +27 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 88%

6. TEST RESULTS

Measurement Date: 2024-08-01
Devices Tested: 96
Insertion Loss (meas): 1.18 dB
Return Loss (meas): 24.4 dB
Rejection (meas): 34.5 dB
Isolation (meas): 44.0 dB
Q Factor (meas): 5664
Group Delay Variation: 4.1 ns
Power Handling: +33 dBm
Die Yield: 95.8%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

a) The Multiplexer Module design demonstrated below-target performance for n78 (3.3-3.8 GHz 5G NR).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Package parasitic effects were consistent with 3D EM model predictions.
e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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